LESSON 8.7

Equations and Inequalities Problem Solving

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can model and solve real-world problems using equations and inequalities.

WHY THIS MATTERS Detective Santos is closing a complex case. She has collected equations and inequalities from different parts of the investigation. One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person $85.' Another says: 'The lookout earned less than $50.' She must model both clues mathematically and check if the answers are reasonable. Can you help?

REMEMBER

Equations and Inequalities Problem Solving

Modeling Real-World Scenarios with Equations & Inequalities To model means to write a real situation as math. I use an equation when there is one exact total, and an inequality when there is a limit or a range of allowed values.

1. Define the variable and units in the scenario.
2. Look for keywords: 'equals/is' (=) vs 'at least/limit' (≥, ≤, >, <).
3. Write the algebraic model, solve with inverse operations, and check in context

Words: **Model *(Modelar)*** **Equation *(Ecuación)*** **Inequality *(Desigualdad)*** **Reasonableness *(Razonabilidad)*** **Variable *(Variable)***

1 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 4 < 11$

(A) $x < 7$

(B) $x > 15$

(C) $x = 15$

(D) $x < 15$

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: $x + 6 > 14$

(A) $x < 8$

(B) $x = 8$

(C) $x > 8$

(D) $x > 20$

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: $x + 10 \geq 25$

(A) $x \geq 15$

(B) $x \leq 15$

(C) $x = 15$

(D) $x \geq 35$

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 9 \leq 3$

(A) $x \leq 6$

(B) $x \geq 12$

(C) $x \leq 12$

(D) $x = 12$

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A detective has a $500 budget for equipment. She has already spent $320. The inequality 320 + r ≤ 500 represents the remaining amount r she can spend. She also needs exactly 4 flashlights at $15 each, represented by the equation 4f = 60. Solve 320 + r ≤ 500.

(A) r ≤ 180
(B) r ≥ 180
(C) r ≤ 500
(D) r ≤ 820

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective has a $500 budget for equipment. She has already spent $320. The inequality 320 + r ≤ 500 represents the remaining amount r she can spend. She also needs exactly 4 flashlights at $15 each, represented by the equation 4f = 60. Solve 4f = 60.

(A) f = 15
(B) f = 56
(C) f = 64
(D) f = 240

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether a word problem needs an equation or an inequality?

SENTENCE STARTER I use an equation when the problem says ___, but I use an inequality when the problem says ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.7

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WHY THIS MATTERS Detective Santos is closing a complex case. She has collected equations and inequalities from different parts of the investigation. One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person $85.' Another says: 'The lookout earned less than $50.' She must model both clues mathematically and check if the answers are reasonable. Can you help?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Room 1: The case board is a mess. Sort each clue into whether it needs an EQUATION (one exact total) or an INEQUALITY (a limit or range).

Groups: Needs an Equation (=) | Needs an Inequality (<, >, ≤, ≥)

_____ The ransom was split equally among 5 suspects, giving each $40.

_____ A number of fingerprints plus 9 equals 21.

_____ The getaway car held no more than 6 bags of loot.

_____ The alarm must stay on for at least 30 minutes.

_____ The witness saw fewer than 4 accomplices.

_____ Three identical safes hold exactly 75 coins in all.

2 **Write the letter of the matching item on each line.**

Room 5: Match each crime-scene clue to the model the detective should write.

_____ The reward divided among 4 detectives gave each $50.

_____ A number of clues take away 7 leaves exactly 9.

_____ The patrol must last more than 8 hours.

_____ The locker holds at least 25 files.

(A) $4d = 200$, so $d = 50$

(B) $c - 7 = 9$, so $c = 16$

(C) $h > 8$

(D) $f \geq 25$

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A detective has a $500 budget for equipment. She has already spent $320. The inequality 320 + r ≤ 500 represents the remaining amount r she can spend. She also needs exactly 4 flashlights at $15 each, represented by the equation 4f = 60. Why is one an equation and the other an inequality?

(A) Because equations are for money and inequalities are for counting

(B) Because 4 is smaller than 320

(C) There is no real difference

(D) The flashlights cost exactly $60, but the budget is a limit she must stay under

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Room 2: The vault keypad clue reads, 'A number of stolen coins plus 6 equals 14.' Solve x + 6 = 14 to find the code x.

(A) x = 2

(B) x = 6

(C) x = 8

(D) x = 20

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Room 3: A sign warns, 'The safe holds at most 20 gold bars.' Which inequality models the number of bars x?

(A) x ≤ 20

(B) x ≥ 20

(C) x > 20

(D) x = 20

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Room 4: A locked drawer clue reads, 'Five identical evidence boxes weigh 60 pounds in all.' Solve 5w = 60 for each box's weight w.

(A) w = 12

(B) w = 55

(C) w = 65

(D) w = 300

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether a word problem needs an equation or an inequality?

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